Anyflo DeFi Provider

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Why we are building Anyflo

Businesses want to pay invoices in fiat, not operate wallets and DeFi. Cross-border payments can be slow, opaque and difficult to reconcile.

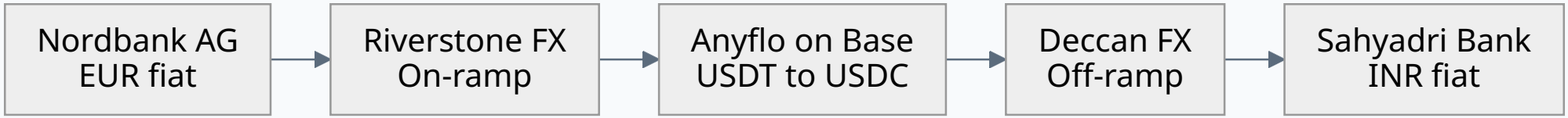
- Keep familiar bank payments at both ends.
- Use stablecoins and on-chain liquidity for the international routing leg.
- Give banking and ramp partners a traceable payment record to review.

Product vision: fiat in, fiat out. This demo covers the on-chain leg.

User story: Atlas pays Deccan

Atlas Motors GmbH in Munich owes *Deccan Parts Pvt. Ltd.* in Pune EUR 50,000 for invoice DP-2048.

Atlas wants to fund in EUR. Deccan wants INR in its bank account.



Illustrative scenario: names are not claimed customers or partners. Fiat rails are product vision, not a demonstrated integration.

A bank-reviewable trail for invoice DP-2048

Target audit record — future work, not implemented in this demo:

- Business identities, invoice purpose, source-of-funds evidence and screening decisions.
- Proposed route, wallets, tokens, pools and venues; timestamped policy version and pre-bid decision, rechecked before execution.
- Linked EUR bank reference, ramp conversion, deposit/swap/payout receipts and INR bank-payout confirmation.

This supports bank due diligence; it does not guarantee acceptance.

The Solver: from inventory to liquidity access

A Solver is the liquidity and execution agent, not a new payment network.

The existing Solver needs payout inventory. This project adds a standalone agent that uses verified customer funding to obtain the payout asset through Uniswap AMMs or a custom Aqua position.

One configured EVM pair, fixed input, one strategy and one active order.

Inventory: the pre-funded baseline

Advantages: no DEX quote or swap on the payment path; fewer external dependencies and more predictable execution timing.

Disadvantages: the Solver must prefund payout assets on destination chains, monitor balances and rebalance. Capital can sit idle or be unavailable where the next payment needs it.

Best fit: frequent, predictable corridors with well-funded inventory.

Uniswap: pooled liquidity, execution trade-offs

Advantages: access LP-funded pools through API routing without the Solver stocking every payout asset. Quotes and receipts expose the route.

Disadvantages: pair/size availability, price impact, pool fees and gas. Quotes do not reserve liquidity; API/RPC latency and Permit2/router validation add dependencies and integration work.

Best fit: same-chain payments with sufficient, policy-acceptable pools.

Aqua: programmable liquidity, backing matters

Advantages: custom position economics on SwapVM; makers retain wallet custody and can share backing across app allocations.

Disadvantages: a usable maker position must exist for the pair/size. Virtual allocations are not locked funds: wallet balances and allowances can change. NO SANDBOX.

Best fit: configured makers needing programmable liquidity positions.

What we built for this hackathon

- Public standalone Solver: existing JWT, signed bids/acceptance and CBOR API.
- Uniswap API quotes, bounded Permit2 authorization and validated AMM swaps.
- Permissionless Aqua/SwapVM position app, maker lifecycle tools and execution.
- Durable spend checkpoints and a runnable mock-Hub, real-contract demo.

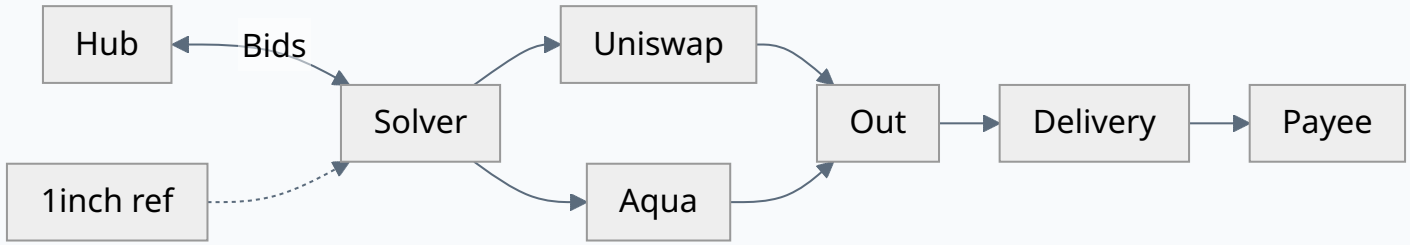
Why compliance must cover the route

Payment providers may need to meet sanctions, AML and internal risk obligations. Screening the sender and recipient alone does not assess how funds move.

- An AMM route can involve multiple tokens, pools and venues.
- A payment provider needs an auditable route decision consistent with policy.
- Requirements depend on jurisdiction, customer and applicable law.

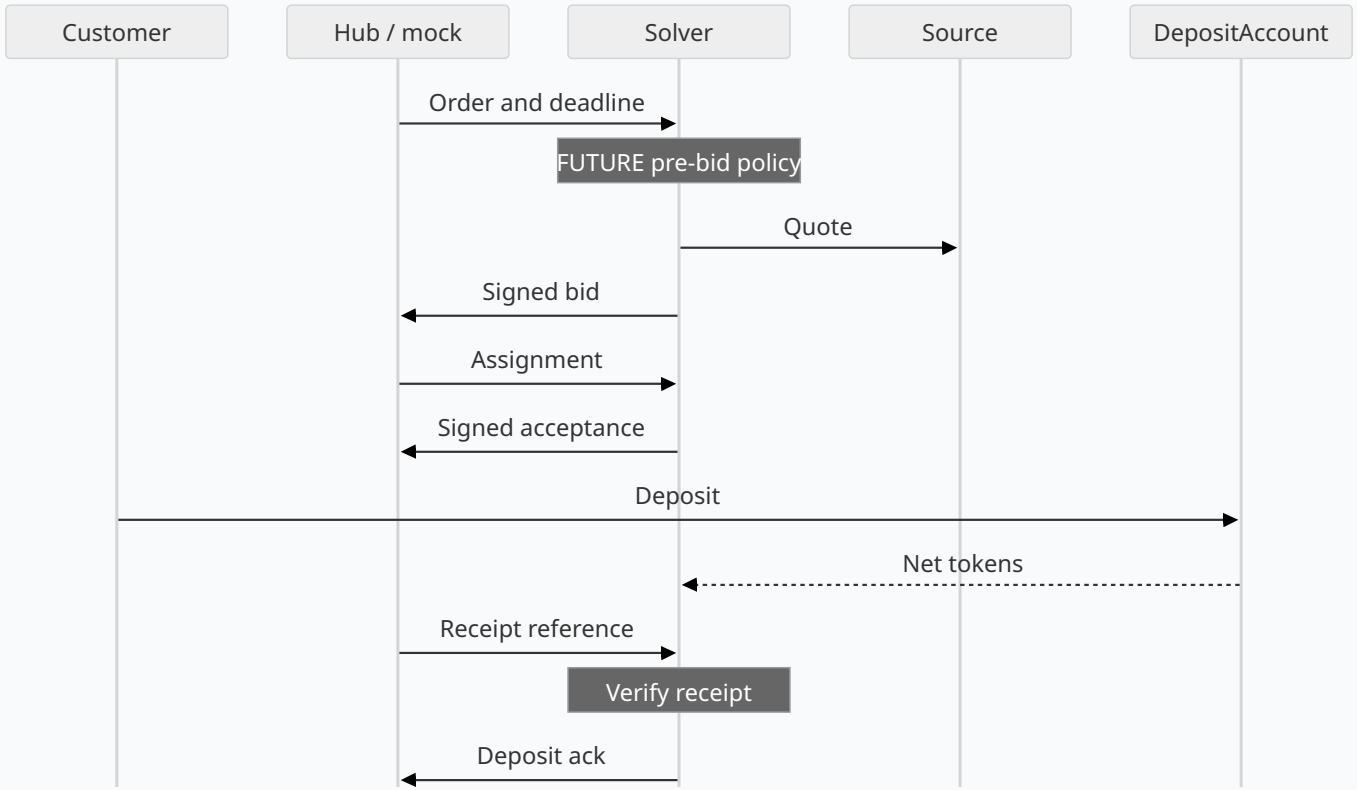
Today: Anyflo screens counterparties; Solver route checks are future work.

Integration: choose a source, preserve Anyflo settlement

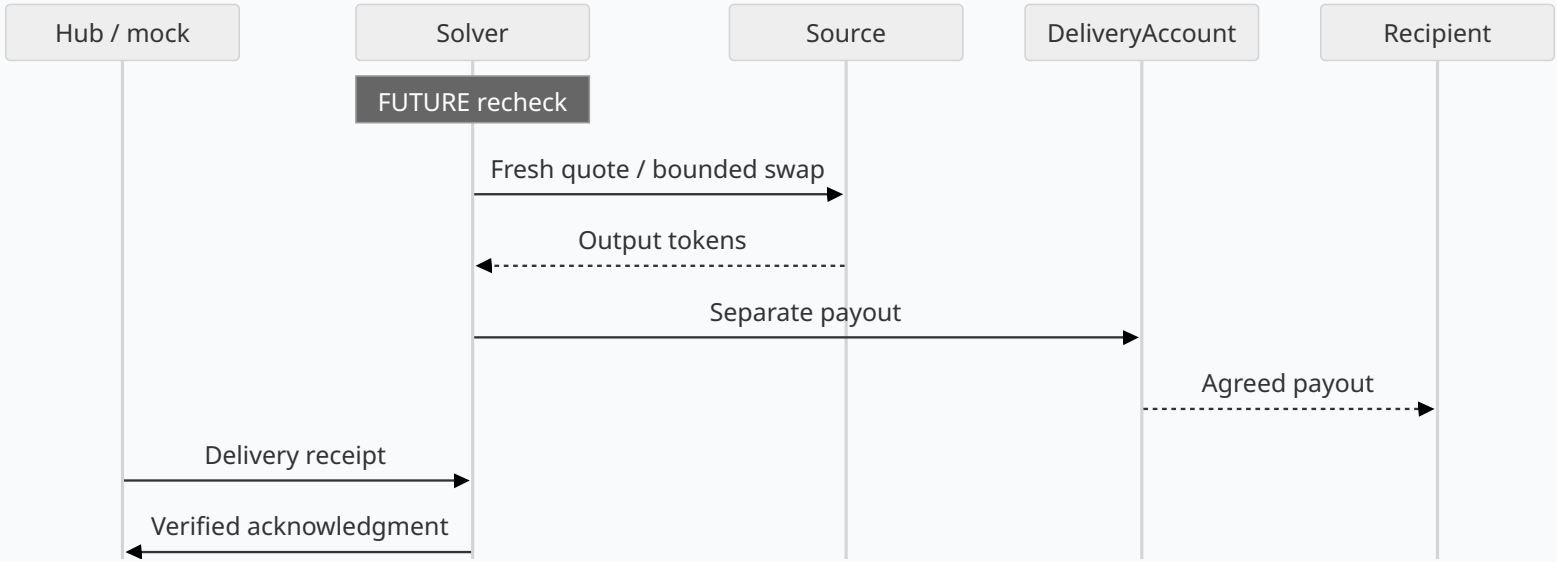


Aqua's executable quote is on-chain. The 1inch HTTP quote is reference data. No resolver enrollment, global position crawler or provider fallback.

Swimlane: auction and funding



Swimlane: execution and delivery



Accepted payout stays fixed. Swap and delivery are separate transactions.

Demo: two deposit-funded payments

```
bun run demo --strategy both
```

Two independent orders on Anvil (31337), using minted test USDT/USDC: one via Uniswap, one via Aqua.

1. The mock Hub publishes an order; a real Solver process signs its bid and acceptance.
2. A verified customer deposit funds the actual AMM/SwapVM swap; the Solver starts with zero token inventory.
3. A separate payout delivers USDC. The report checks receipts, fees, premium and recipient balances; Aqua also proves ship/fill/dock.

Mocked: Hub, provider HTTP and Anyflo settlement contracts. No fiat rails, mainnet or real funds.

Aqua app: a wallet-backed programmable position

Position: 0.99–1.01 band, 5 bps input fee, 30-second decay.

- A separate maker ships a permissionless position on the official SwapVM engine.
- Virtual reserves refer to actual maker-wallet balances and registry allowances.
- Real ship, fill and dock receipts were observed; docking leaves custody intact.

Observed local payment results

Strategy	Input	Swapped	Delivered USDC	Checks
Uniswap V3	10.000000	9.990000	9.934087	11
Aqua / SwapVM	10.000000	9.990000	9.934580	14

Both start with zero Solver token balances. Separate protocol fee: 0.100000 test USDT; retained input premium: 0.010000. These are one local run, not rates.

What worked well

- Published Anyflo wire types and helpers let the public Solver reuse the existing API without private backend source.
- Both real local engines swapped deposit-funded tokens; the Solver started with no input/output inventory.
- Receipts proved fee separation, exact customer payout and Aqua ship/fill/dock.
- Focused guards covered unsafe spending, short funding and duplicate delivery.

Real-world simulation: USD 250k to AED

Transfer: USD 250,000. Recipient currency: AED (Emirati dirham).

Payment approach	Fees (USD)	Fees / principal
Banking route	12,000	4.800%
Inventory (existing)	4,000	1.600%
Uniswap	2,680	1.072%
1inch Aqua	2,610	1.044%

Totals include estimated on/off-ramp costs where applicable. These are supplied scenario estimates, not live quotes or protocol tariffs.

What did not work as first assumed

- Live quotes took roughly 2–4 seconds: too slow for a 1.5-second auction; the demo used 10 seconds.
- A V3-only Base request found no route; enabling V2/V3/V4 produced a usable AMM quote.
- Permit2/wire/ABI assumptions and a launcher hiding compiler failure caused integration defects; these were corrected.
- Hosted APIs cannot quote Anvil-only tokens: local HTTP mocks are necessary, not live-provider proof.

Developer experience: what should improve

- **Version contracts:** a maintained API, Permit2/router and Aqua/SwapVM SDK/ABI compatibility matrix.
- **Complete examples:** quote, fund, bounded authorization, swap, receipt proof; explain permissionless Aqua and both approval directions.
- **Repeatable fixtures:** official test-token deployments and reliable CLI failure codes, rather than bespoke compiler glue.
- **Operability:** typed failure reasons and reconciliation guidance without exposing credentials.

Added value

Anyflo and business customers: more liquidity paths without stocking every payout asset in each Solver wallet; the existing order API stays unchanged.

Uniswap and Aqua: business-originated payment flow and a potential use case connecting DeFi liquidity to regulated payment rails.

LP capital, execution risk and applicable legal obligations remain.

Compliance roadmap — explicitly not built here

Existing production context: Anyflo access controls and counterparty screening. The mock demo does not execute those backend checks.

Future Solver goal: inspect the complete execution plan, every route, asset, venue and counterparty **before bidding and again before executing**. A changed quote or route must be checked again.

Transaction-bound validation is implemented; **route AML/policy enforcement is not**.

Evidence and boundaries

Verified: 43 Bun tests, 5 native Forge settlement tests, both real local protocol flows, and live read-only API probes. Source pins were also verified.

Mocked: Hub/customer role, local provider HTTP and Anyflo settlement doubles. **Not verified:** production backend integration or mainnet settlement.

These observations do not establish a production SLA, a best-price guarantee or banking/compliance acceptance.

Next steps: from prototype to bank-facing payments

1. **Production integration:** integrate tested Uniswap/Aqua adapters alongside existing inventory, with reservations, monitoring and receipt-based recovery.
2. **Policy and evidence:** implement pre-bid/requote checks and the audit trail; review requirements with banking/ramp partners.
3. **Controlled pilot:** clear security/license reviews, verify deployments and liquidity, then authorize a capped mainnet test.
4. **Fiat edges:** integrate eligible ramps, FX/fee quotes and payout confirmations against the same invoice record.